Data Speaks!! — Music Tribes Explorer

1 Music Tribes Explorer: clustering and dimensionality reduction

1.1 Goal and setup

The question this module answers is what the natural groupings of music are when nothing but sound is allowed to define them. No genre label, artist, year or popularity figure enters the model; the human labels are held back and used afterwards only to check the result.

The input is the project's frozen, preprocessed sample of 100,001 tracks and the shared `StandardScaler`, loaded and never re-fitted. Standardisation is not optional here: loudness spans roughly −60 to 0 dB and tempo 50 to 200 BPM, so on raw values a Euclidean distance would be almost entirely a tempo distance. Every result below uses `SEED = 42`.

1.2 Choosing k

K-Means was fitted for $k = 2 \dots 15$ with `n_init=10`. Silhouette is $O(n^2)$, so it is evaluated on a fixed 10,000-track subsample; inertia, Davies–Bouldin and Calinski–Harabasz use all 100,001 tracks.

k	Inertia ($\times 10^3$)	Silhouette	Davies–Bouldin	Calinski–Harabasz	Smallest cluster
3	895.5	**0.214**	1.710	16752	5047
4	830.6	0.183	1.783	14645	4998
5	770.6	0.156	1.909	13789	4937
6	726.1	0.161	1.788	12930	4820
7	690.8	0.161	1.768	12177	4706
8	659.0	0.156	1.652	11630	1268
9	628.0	0.154	1.632	11296	307

Table 1: Model selection sweep on the 12 standardized audio features. The full sweep ($k = 2 \dots 15$) is in `k_sweep_12feature.csv`.

The two criteria disagree, and the disagreement is worth stating plainly rather than hiding. Silhouette is maximised at $k = 3$; the elbow falls at $k = 6$. Inspecting the $k = 3$ solution shows why it scores well: it separates acoustic, loud-energetic and instrumental music and stops there. It is a defensible clustering and a useless explorer, because a website legend with three entries called *quiet*, *loud* and *no vocals* tells a visitor nothing they could not hear in five seconds.

At the other end, $k \geq 8$ degenerates. At $k = 8$ a 1,268-track cluster appears whose defining property is `time_signature` $= 0$, a Spotify encoding artifact rather than a musical fact; from $k = 9$ the smallest cluster collapses to about 300 tracks. $k = 6$ **is committed**: it is the elbow, it ties $k = 7$ on silhouette (0.1610 vs 0.1611), and it is the largest k at which no cluster is an artifact.

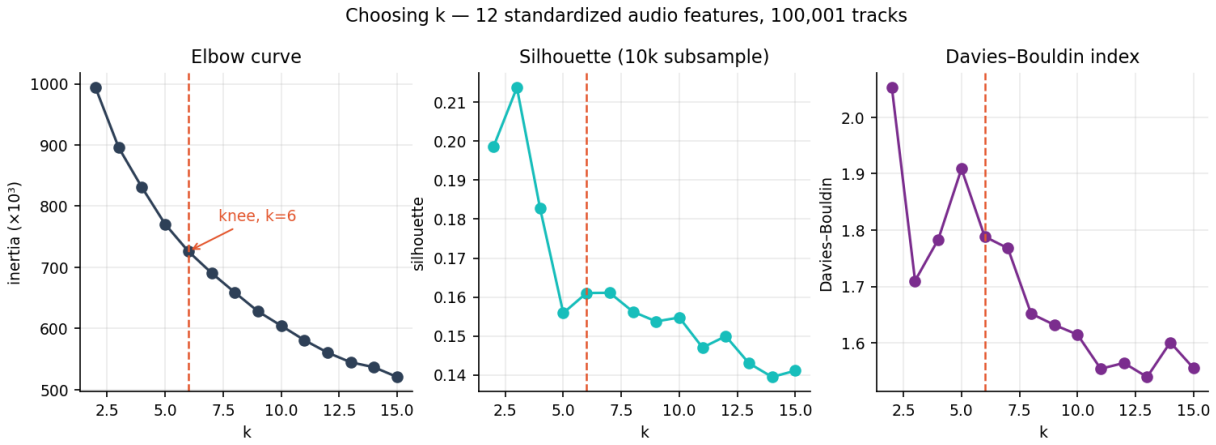


Figure 1: Elbow, silhouette and Davies–Bouldin against k. The dashed line marks the chosen $k = 6$; the elbow is located as the point furthest from the chord joining the endpoints of the inertia curve.

Absolute silhouette values are low (0.16) throughout. This is expected and is itself a finding: recorded music does not form isolated balls in audio-feature space, it forms a continuum. The clusters are regions of a gradient, not islands, which is exactly what the UMAP map in Figure 5 shows.

1.3 The six tribes

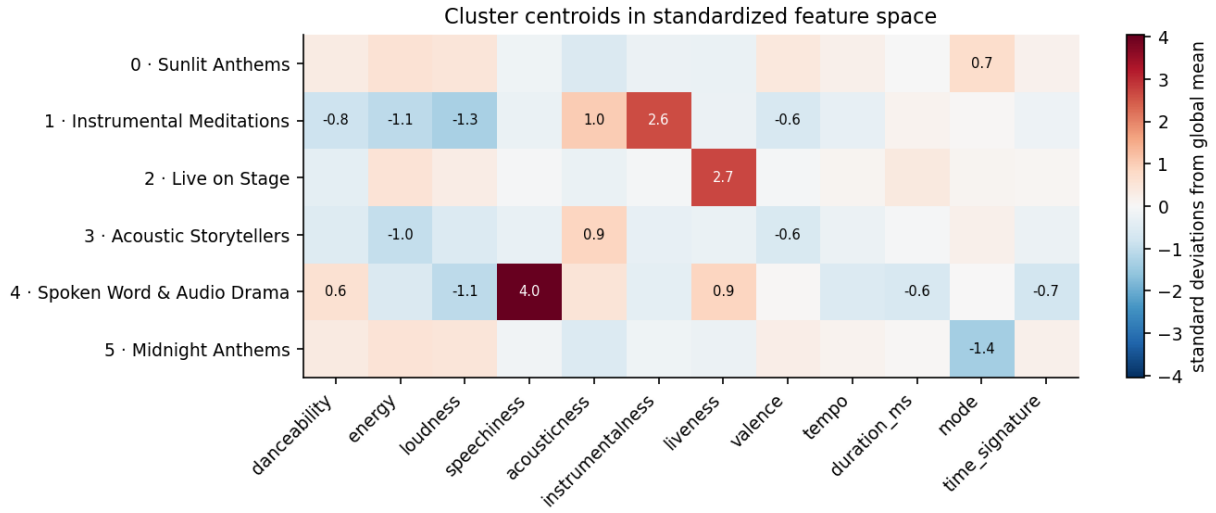


Figure 2: Cluster centroids in standard deviations from the global mean. Each tribe is defined by one or two dominant features.

Names were assigned from the centroid profiles and the per-cluster artist and song tables, and only then checked against genre labels.

0 – Sunlit Anthems is the loud, upbeat, full-band end of the corpus: classic rock, album rock and soft rock alongside a large Latin and *filmi* contingent. **1 – Instrumental Meditations** is driven by a single feature at +2.64 SD, instrumentalness; Bach, Mozart, Chopin and Beethoven sit here with cool jazz and bebop. **2 – Live on Stage** is separated by liveness at +2.69 SD. It is a recording context rather than a sound, and it is named literally to avoid implying otherwise. **3 – Acoustic Storytellers** pairs high acousticness with near-zero instrumentalness, meaning

ID	Name	Tracks	Share	Defining profile (raw units)	Sil.
0	Sunlit Anthems	33,914	33.9%	energy 0.69, loudness −7.7 dB, valence 0.67, acousticness 0.25, all major key	0.19
1	Instrumental Meditations	8,611	8.6%	instrumentalness 0.81, loudness −17.0 dB, acousticness 0.81	0.09
2	Live on Stage	6,500	6.5%	liveness 0.71, energy 0.68, longest tracks (4:39)	0.07
3	Acoustic Storytellers	24,504	24.5%	acousticness 0.75, energy 0.30, valence 0.39, instrumentalness 0.02	0.13
4	Spoken Word & Audio Drama	4,820	4.8%	speechiness 0.87, duration 2:27, irregular metre	0.23
5	Midnight Anthems	21,652	21.7%	energy 0.68, loudness −7.7 dB, valence 0.62, all minor key	0.16

Table 2: The six tribes. Silhouette is the mean per-cluster value on a 20,000-track subsample.

acoustic backing plus a voice: Sinatra, Ella Fitzgerald, Billie Holiday, Lata Mangeshkar. **4 – Spoken Word & Audio Drama** is the cleanest cluster in the model (speechiness +4.04 SD) and is not music at all: German *Hoerspiel* series, Scandinavian *barnsagor*, audiobook chapters. **5 – Midnight Anthems** carries the same energy, loudness and danceability as cluster 0 but in minor keys; it skews later (median 2001 vs 1998) and holds more hip-hop (14% vs 9%) and electronic music. Section 1.7 explains why this pair exists.

1.4 Validation against human genre labels

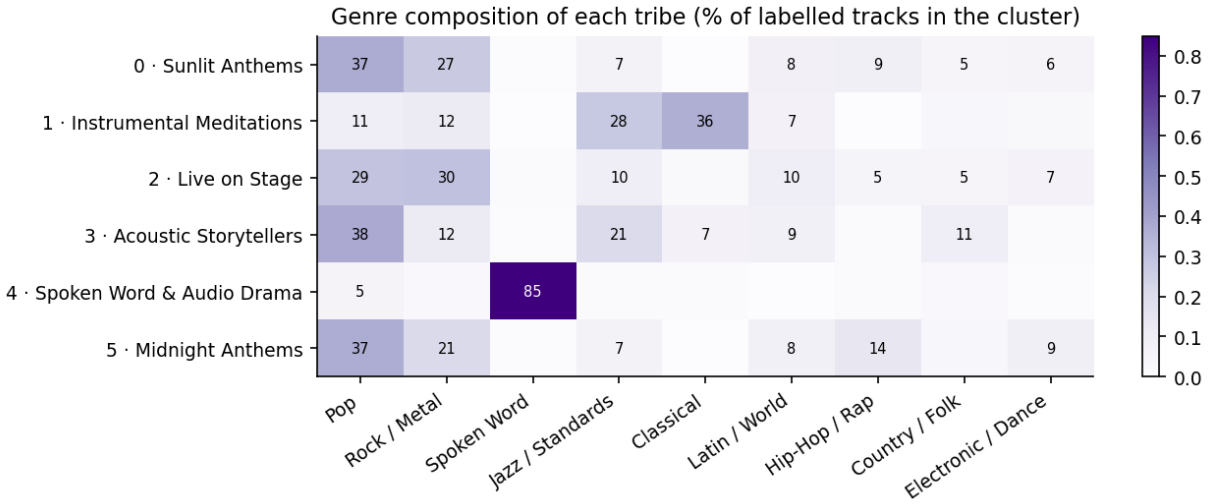


Figure 3: Genre composition of each tribe. Spotify's raw genre strings are too fine-grained to tabulate directly, so artist genres are mapped by keyword into nine coarse buckets; 77.2% of tracks receive a bucket.

The plan proposed a threshold test: if an unsupervised cluster is 70% or more one genre, the clustering has rediscovered genre structure from sound alone. **One cluster passes.** Spoken Word & Audio Drama is 84.9% spoken-word labels. Everything else fails the test: Instrumental Meditations is 36% classical and 28% jazz, and the anthem and storyteller tribes are each a mix of pop, rock, Latin and standards. Overall agreement is weak (NMI = 0.144).

The honest reading is that audio features capture *production and instrumentation*, not genre. A 1965 Beatles track and a 2015 Latin pop track can share energy, loudness and valence while sharing no listener-facing genre; conversely, an acoustic hip-hop track and a trap track sit in different tribes. Genre is a social and historical category that audio descriptors only partially reach. That is a result about the data, not a failure of the method, and it is the most useful thing this module has to say.

1.5 A century of sound

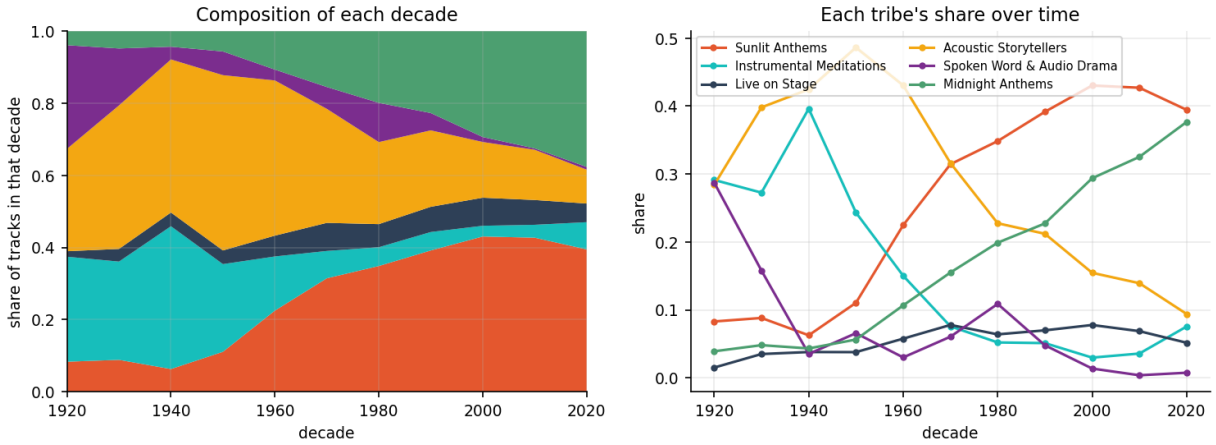


Figure 4: Composition of each decade (left) and each tribe's share over time (right).

Because the sample is stratified by decade, these shares are comparable across eras. Acoustic Storytellers dominate the mid-century (48.6% of the 1950s) and fall to 9.4% by the 2020s. Instrumental Meditations peak in the 1940s at 39.6% and end at 7.6%. The two anthem tribes together rise from 12.2% of the 1920s to 77.1% of the 2020s, and within that rise the minor-key tribe grows fastest, from 3.9% to 37.7% — consistent with the documented drift of popular music toward minor tonality. Spoken Word is bimodal, prominent in the 1920s (28.7%, early recorded readings) and again in the 1980s (10.9%, the German radio-play boom), then almost vanishing.

1.6 PCA versus UMAP

PCA is linear and global. PC1 (24.5% of variance) is the acoustic-to-electric axis — energy, loudness and danceability against acousticness — and PC2 (13.0%) is talk and groove: speechiness, danceability and valence against duration. Two components retain only 37.5% of the variance, and the scatter is correspondingly a single dense cloud in which only Spoken Word detaches.

UMAP preserves local neighbourhoods far better. Measured as trustworthiness at $k = 15$ on a 5,000-track subsample, UMAP scores 0.938 against PCA's 0.784. It is therefore used for the website map, with PCA reported as the linear baseline. The cost is that UMAP distances between well-separated islands carry no quantitative meaning; only neighbourhoods do.

1.7 Sensitivity: what `mode` and `time_signature` cost

Ten of the twelve prescribed features are continuous descriptors. Two are not: `mode` is binary (major/minor) and `time_signature` is a near-constant small integer. After standardisation each carries unit variance — as much weight as energy — and Euclidean geometry cannot distinguish a category from a quantity.

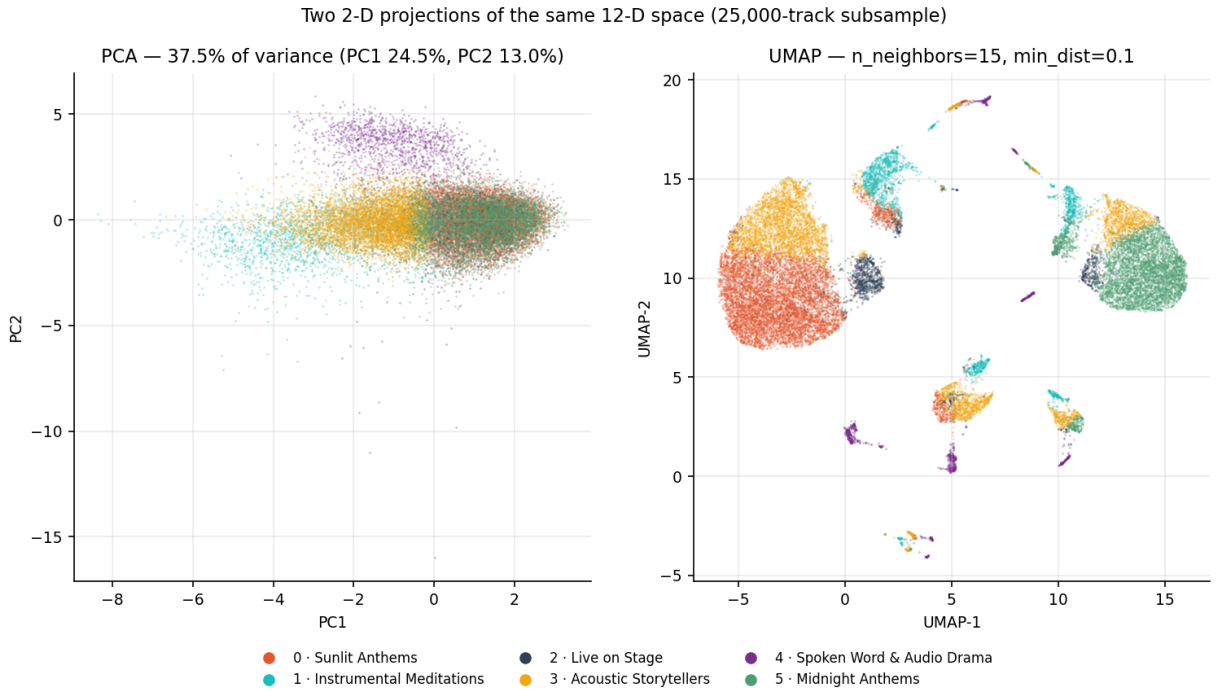


Figure 5: Two projections of the same 12-dimensional space, 25,000-track subsample, coloured by cluster.

The consequences are measurable. Clusters 0 and 5 lie 2.12 apart in the full 12-dimensional space but only 0.232 apart across the ten continuous dimensions, while the next-closest pair of centroids is 2.688 apart: they are one sonic tribe split by key signature, and cluster 0 is 100% major while cluster 5 is 100% minor. The same feature dominates the projection — the first UMAP axis correlates −0.906 with `mode`, dividing the map into a 60,278-track major continent (99.2% major) and a 39,723-track minor continent, each holding a duplicate of every tribe.

Re-running the module on the 10 continuous features, using the same shared scaler and simply selecting ten of its twelve output columns, improves every internal measure: silhouette 0.187 vs 0.161, Davies–Bouldin 1.614 vs 1.788, genre NMI 0.157 vs 0.144, and the UMAP–`mode` correlation falls to −0.017. The two solutions agree moderately (ARI = 0.493); the 10-feature run replaces the major/minor pair with a musically meaningful split between high-energy, fast, loud tracks and slower, high-valence danceable ones.

The 12-feature solution is retained as this module's primary deliverable, because it is the specification in the project plan and because it is the identical feature space in which the DJ Transition Engine builds its k-NN graph, so the map and the graph stay consistent. Both projections are exported in `map_data.json`, so a front end can use either; the 10-feature map is the one where the tribes are legible.

1.8 Limitations

Three, stated honestly. First, the silhouette values are low in absolute terms; these are regions of a continuum, and any claim that a song "belongs" to a tribe is a claim about which region it falls in, not about a natural kind. Second, the clusters answer to production characteristics rather than genre, so the explorer should be read as a map of how records *sound*, not of what they *are*; the 84.9% spoken-word cluster is the one place the two coincide. Third, K-Means assumes isotropic, similarly sized clusters and forces every track into one — there is no outlier class, so the handful of tracks with corrupt `time_signature` values are absorbed into whichever tribe is

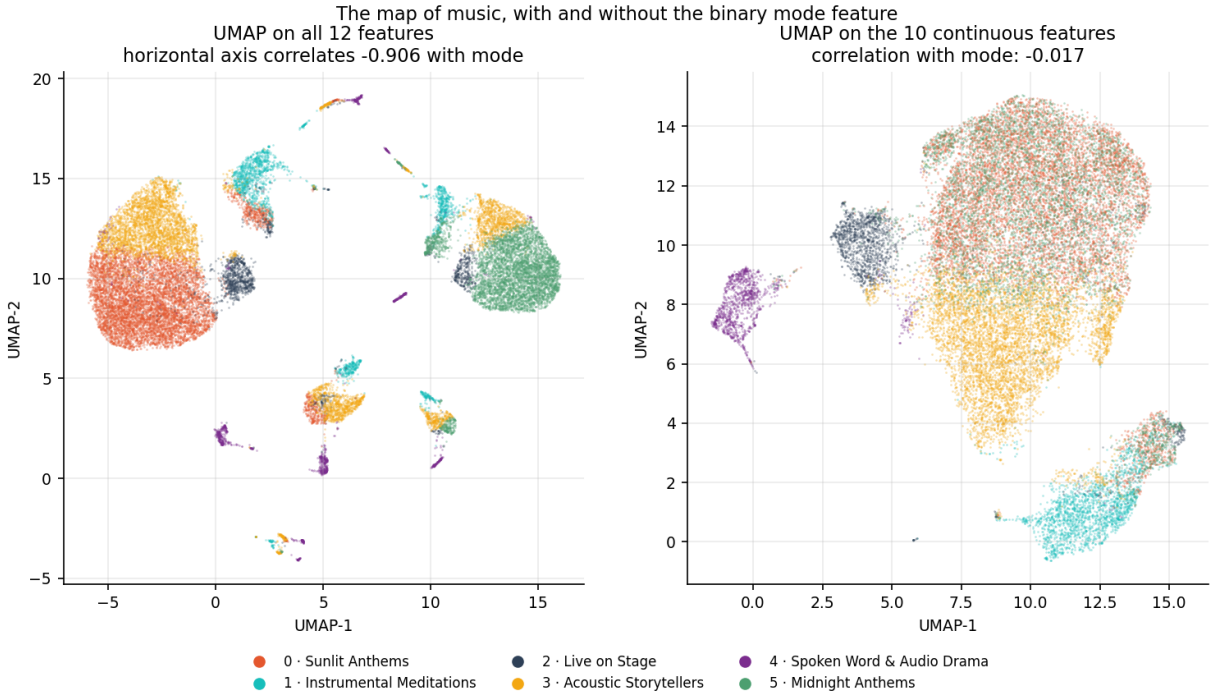


Figure 6: The same K-Means labels on both projections. Left: UMAP on all 12 features, split into mirrored major and minor continents. Right: UMAP on the 10 continuous features, where the twin tribes merge and a single acoustic-to-electric gradient appears.

nearest. A density-based method such as HDBSCAN would leave them unassigned and is the natural extension for the endterm submission.

1.9 Deliverables

`music_tribes.ipynb` (runs end-to-end from a clean kernel); `kmeans.joblib` and `pca.joblib`; `tracks_clustered.parquet` (the 100,001 tracks with `cluster_id`, PCA and both UMAP coordinate pairs); `cluster_metadata.json` (names, profiles, top genres, artists, representative songs, per-decade series); `map_data.json` (4,999 songs for the browser); the CSV tables behind every figure; and this section.